Lightweight Dual Correlation Variance Compensation for Salient Object Detection in Optical Remote Sensing Images

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***Abstract*—Salient object detection in optical remote sensing images (ORSI-SOD) confronts three fundamental challenges: extreme multiscale object heterogeneity where targets span multiple orders of magnitude within a single scene; subtle interclass boundaries with minimal contrast between objects and backgrounds; and computational scalability demands for processing high-resolution imagery. Existing approaches suffer from a critical limitation in that they fail to exploit the discriminative information encoded in interlayer correlation variance, which contains essential boundary and transition cues that are systematically lost during conventional feature fusion processes. This article introduces MambaCVC, a lightweight dual-correlation variance compensation network that explicitly models and compensates for correlation variance patterns to enhance feature discriminability while maintaining linear computational complexity. Our approach centers on two synergistic modules: the global correlation variance compensation module (GC-VCM) that selectively amplifies semantically meaningful long-range dependencies through correlation variance modeling across distant features, and the local correlation VCM (LC-VCM) that preserves fine-grained boundary details via variance-driven local enhancement mechanisms. To achieve computational efficiency, we leverage the VMamba backbone with minimal architectural adaptations optimized for ORSI-SOD. Comprehensive experiments on three public ORSI-SOD datasets demonstrate that MambaCVC achieves state-of-the-art performance with remarkable efficiency, requiring only 6.25G FLOPs and 7.26M parameters.**

***Index Terms*—Correlation variance, feature compensation, lightweight network, mamba, optical remote sensing images (ORSI), salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) represents a fundamental computer vision task that aims to automatically identify regions drawing the most human visual attention [1], [2], with extensive applications in image segmentation [3], [4], [5], object tracking [6], image understanding [7], and scene analysis [8]. While well-established in natural scene contexts, the extension to optical remote sensing images (ORSI) presents fundamentally different challenges that demand specialized methodological innovations.

Received 11 October 2025; revised 11 February 2026; accepted 10 March 2026. Date of publication 13 March 2026; date of current version 7 April 2026. This work was supported by the National Key Projects of China under Grant 2021XJTU0040. *(Corresponding author: Bin Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3673920

ORSI-SOD confronts three interconnected challenges that fundamentally differentiate it from natural scene SOD. First, *extreme multiscale object heterogeneity* manifests as dramatic scale variations where objects span multiple orders of magnitude, from small vehicles occupying 10–20 pixels to large industrial complexes covering 200–500 pixels, often coexisting within a single scene. This heterogeneity creates severe scale imbalances as features from different pyramid levels operate in vastly different representational spaces. Second, *subtle interclass contrast boundaries* present minimal intensity differences between salient objects and backgrounds, particularly in urban or vegetated environments. Unlike natural scenes, remote sensing imagery often contains targets with low signal-to-noise ratios that require enhanced discriminative mechanisms. Third, *dense spatial clustering with ambiguous figure–ground relationships* creates complex spatial arrangements where similar objects form intricate patterns with unclear boundaries, introducing high intraclass variance across diverse environmental conditions.

Traditional handcrafted feature-based methods [9], [10], [11] fundamentally fail in ORSI contexts due to their inability to capture complex statistical dependencies inherent in remote sensing data. While CNN-based approaches [12], [13], [14] have demonstrated improvements, their limited receptive fields create bottlenecks in modeling long-range spatial dependencies critical for distinguishing salient objects from visually similar background regions. Transformer architectures [15] offer global context modeling through self-attention mechanisms [16], yet encounter computational bottlenecks with high-resolution images [17], [18], and critically, oversmoothing of local details that leads to poor boundary localization essential for remote sensing applications.

Recent hybrid CNN–Transformer architectures [19], [20] attempt to combine local feature extraction with global modeling, yet suffer from *semantic misalignment across hierarchical levels* where global and local features operate in different representational spaces. More critically, existing methods systematically ignore *interlayer correlation variance*, the statistical variations in feature correlations between different scales that contain crucial discriminative information for boundary detection and object localization.

In addition to ORSI-SOD, small target detection in infrared imagery has also attracted extensive attention in recent years due to its importance in surveillance and early warning

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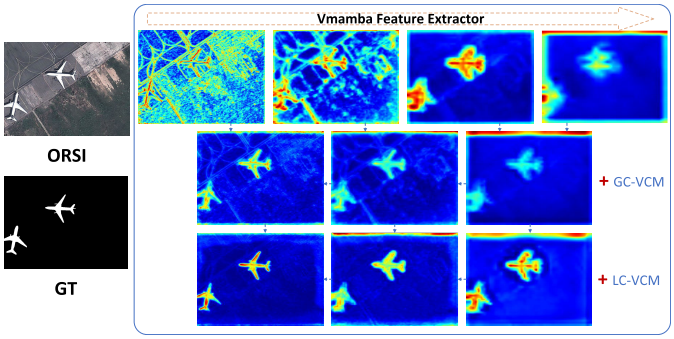


Fig. 1. Visualization of the MambaCVC feature extraction process. (Top row) Raw VMamba feature maps exhibiting diffuse activations with significant background interference and poor target localization. (Middle row) Improved global context modeling after applying GC-VCM, showing more focused activations on aircraft targets while suppressing background correlations. (Bottom row) Final enhancement after applying LC-VCM, where sharp, well-defined feature responses closely align with actual aircraft boundaries.

systems. Representative works [21], [22] enhance infrared small target detection by exploiting multibranch feature interaction and explicit edge-aware guidance, respectively. These methods demonstrate that explicitly modeling structural cues, boundary information, and interfeature interactions is crucial for detecting small targets under low contrast and cluttered backgrounds.

Although infrared small target detection and ORSI-SOD differ in sensing modality and data characteristics, they share common challenges, including small object size, weak object–background contrast, and ambiguous boundaries. The success of these methods further motivates the exploration of principled mechanisms to enhance discriminative feature representations and boundary sensitivity in ORSI-SOD.

Our investigation reveals a fundamental insight illustrated in Fig. 1: interlayer correlation variance encodes critical discriminative information that existing methods systematically suppress. Through extensive analysis, we observe that regions exhibiting high correlation variance consistently correspond to semantically important boundaries and object transitions, while low-variance regions represent homogeneous backgrounds. This correlation variance serves as a natural discriminative signal, representing uncertainty in interlayer feature relationships where high-variance regions indicate inconsistent correlations typically occurring at semantic boundaries.

The visualization in Fig. 1 demonstrates this principle through three progressive stages. Raw VMamba features exhibit diffuse activations with poor object–background discrimination and significant background interference. Our global correlation variance compensation module (GC-VCM) produces significantly improved global context modeling with enhanced focus on target regions while suppressing background interference. The local correlation VCM (LC-VCM) further refines features into sharp, well-defined responses that accurately align with object boundaries, validating that explicit modeling of interlayer correlation variance enables superior discriminative feature learning.

Recent advances in state-space models (SSMs), particularly Mamba [23], offer promising solutions for efficient global modeling with linear complexity. The VMamba architecture [24] provides efficient long-range dependence modeling essential for handling spatial scale heterogeneity in ORSI. However, direct application reveals specific limitations: the blockwise modeling approach lacks sufficient cross-boundary information flow, leading to blurred object contours and compromised fine structural details. This becomes particularly severe when detecting small salient targets against cluttered backgrounds, where boundary preservation is crucial.

To address these ORSI-specific challenges, we propose MambaCVC, a lightweight network centered on a novel *dual-correlation variance compensation* strategy that explicitly models and compensates for correlation variance patterns to enhance feature discriminability while maintaining computational efficiency. The GC-VCM addresses multiscale heterogeneity through cross-scale attention mechanisms and correlation variance modeling across distant features. Through its GC transfer unit (GCTU) and variance compensation unit (VCU), GC-VCM selectively amplifies semantically meaningful long-range dependencies while suppressing redundant correlations from homogeneous backgrounds. The LC-VCM tackles subtle contrast and boundary ambiguity through local variance compensation mechanisms, improving sensitivity to subtle intensity differences while preserving fine boundary details essential for accurate object delineation.

Furthermore, we adopt VMamba as our backbone architecture due to its linear complexity advantages and global modeling capabilities. We incorporate minimal but strategic modifications to the stem module to better preserve spatial details critical for remote sensing applications, while maintaining the backbone's computational efficiency.

Our primary contributions are as follows.

1) We propose a novel dual-correlation variance compensation framework that explicitly models and compensates for interlayer correlation variance to enhance feature discriminability, addressing semantic misalignment in multiscale feature fusion for ORSI-SOD.
2) We develop the GC-VCM with principled correlation variance modeling across distant features via GCTU and VCU components, specifically designed to handle complex spatial arrangements and scale variations in remote sensing imagery.
3) We design the LC-VCM that improves local feature discrimination through theoretically grounded variance-driven enhancement mechanisms, targeting low interclass contrast and boundary ambiguity challenges inherent in ORSI.
4) We demonstrate that leveraging the VMamba backbone with our correlation variance compensation strategy achieves superior efficiency–accuracy tradeoffs compared to CNN and Transformer alternatives through comprehensive experimental validation.

II. Related Work

A. CNN-Based SOD in ORSIs

CNN-based ORSI-SOD explores the unique properties of ORSI data and has made great progress. LVNet [25] uses

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dual-flow pyramid modules and nested structures to accurately localize salient targets at different scales. DAFNet [26] densely integrates low-level features into the deeper layers and incorporates attentional mechanisms to guide feature learning. PDFNet [27] utilizes semantic guidance and parallel attention fusion to refine the target. MJRBM [28] uses a joint learning approach to optimize both boundary and multiscale features of salient objects. EMFI-Net [29] performs salient target detection through multiscale feature integration and exploitation of edge information. ACCoNet [30] coordinates and explores adjacent contextual information to capture multiscale content. MCCNet [31] models the complementary nature of multiple contents of ORSI data and can highlight salient targets well. SEINet [32] employs the interaction between the two branches of edge and semantics, fully combining the advantages of both attentional and edge-aware mechanisms. AESINet [33] adaptive edge-aware semantic interactions for effective salient target detection. MCP-Net [34] leverages cross-image contextual information through a memory-based context propagation network to enhance ORSI-SOD by incorporating dataset-level and scale-aware contexts. WeightNet [35] minimizes noise and enhances feature fusion and target localization across scales by employing a multiscale, multidimensional weighted network. RAGRNet [36] enhances ORSI-SOD in complex contexts by integrating region and boundary features through adaptive graph inference for accurate and complex object modeling. However, these methods can still struggle with accurately capturing the complexity and variability of salient objects in ORSI, and often face challenges in balancing the trade-off between model complexity and detection performance. In addition, while lightweight models [37], [38], [39] offer efficiency, they may compromise on the depth and breadth of feature learning necessary for complex ORSI scenarios.

B. Transformer-Based SOD in ORSIs

To overcome the shortcomings of CNNs in long-distance dependence modeling, the Transformer model is introduced to the ORSI-SOD task. Transformer is able to capture global features efficiently through the self-attention mechanism, and is especially good at dealing with long-distance dependencies in images. A novel transformer framework (ASTT) [40] for ORSI-SOD overcomes CNNs limitations in capturing long-range relationships by leveraging global–local feature extraction with an adaptive spatial tokenization encoder and a dense token aggregation decoder. GeleNet [17] introduces a novel global-to-local paradigm for ORSI-SOD by leveraging a transformer backbone and incorporating attention mechanisms to enhance local interactions and cross-level contextual information. BMGT [41] enhances ORSI-SOD by leveraging a bidirectional mutual guidance approach between global context-aware representations and fine-grained boundary structures, utilizing a hierarchically structured transformer encoder and a dual-stream MLP decoder for comprehensive feature interaction. TLCKD-Net [18] introduces a transformer architecture for ORSI-SOD with a large convolutional kernel decoding network to effectively capture long-range dependencies and multiscale features.

C. Hybrid CNN–Transformer Models for SOD in ORSIs

Combining the local feature extraction capability of CNN and the global modeling capability of Transformer, some hybrid models have been proposed to enhance the performance of ORSI-SOD. GLSIN [19] is an effective approach for ORSI-SOD that leverages scribble supervision, employing a dual-branch encoder combining a Transformer and CNN to capture global and local features while enhancing feature interaction through specialized blocks. GSANet [42] is a global semantic-aware aggregation network designed to effectively prioritize regions with high information entropy and enhance feature aggregation through modules that fuse shallow and deep features. HFANet [43] is a hybrid feature-aligned network designed for ORSI-SOD, effectively addressing challenges like cluttered backgrounds and scale variation by combining global context modeling, a Gated Fold-ASPP for multiscale feature extraction, and an interactive guidance loss to enhance saliency and edge detection. DCPNet [44] is a dual-stream feature collaboration perception network that enhances ORSI-SOD by integrating CNN and Transformer, employing a dual-branch feature extractor for multiscale extraction, a multipath complementary-aware interaction module for fine-grained feature fusion, and a feature weighting balance module to ensure a balanced focus on both global and local features. HFCNet [45] is a heterogeneous feature collaboration network for ORSI-SOD, effectively addressing challenges like irregular object topology and cluttered backgrounds by integrating CNN and Transformer features, utilizing an adaptive global–local integration module, and employing a saliency-guided attention-enhanced decoder to focus on important areas while suppressing irrelevant information.

D. Mamba

In recent years, Mamba [23] has emerged as a powerful alternative to CNNs and Transformers for visual modeling, owing to its ability to capture long-range dependencies with linear computational complexity. It achieves efficient sequence modeling by leveraging an SSM, and has shown promising results across a variety of vision tasks [46]. Recent extensions have broadened its applicability to remote sensing tasks [47]. For instance, RSMamba [48] enhances the Mamba backbone with a dynamic multipath activation mechanism to improve noncausal sequence modeling in remote sensing image classification, yielding state-of-the-art performance on several benchmark datasets. RSM [49] targets dense prediction in very-high-resolution remote sensing imagery by incorporating an omnidirectional selective scan module to improve global context modeling while maintaining linear complexity. CVMH-UNet [50] and LoG-VMamba [51] integrate visual state-space modeling into CNN-based segmentation networks to achieve accurate results on satellite and medical images while preserving computational efficiency. Moreover, CDMamba [52] introduces a global–local Mamba framework for bitemporal remote sensing change detection, combining convolution-enhanced local detail modeling with adaptive global–local feature interaction. Beyond optical imagery, SSMamba [53] explores spectral–spatial token generation and

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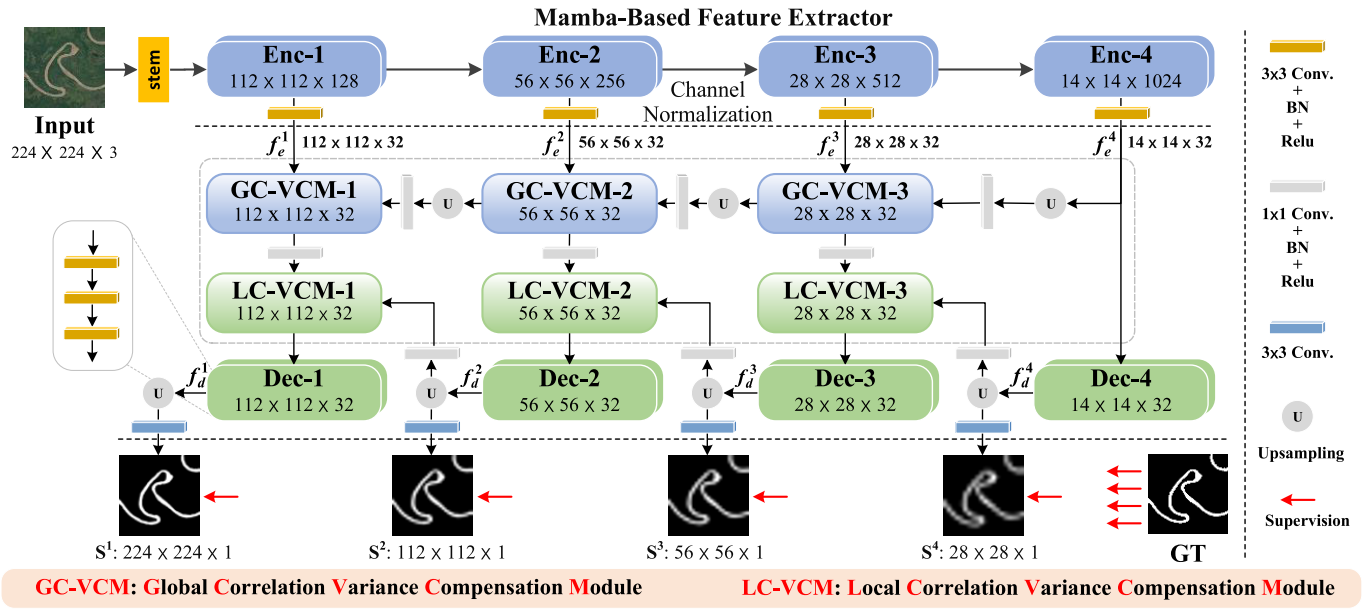


Fig. 2. Overall framework of MambaCVC with feature extraction, interaction, and enhancement. First, multiscale features are extracted using VMamba-Base as the backbone network and are processed with unified channel dimensions. Next, these multiscale features are fed into the GC-VCM and LC-VCM to facilitate feature interaction and enhancement. Finally, the decoder progressively infers the salient objects. During the training phase, a scale-matching supervision strategy is applied to all decoder blocks to ensure consistency and accuracy across different feature scales.

stacked Mamba blocks for hyperspectral image classification, further demonstrating the flexibility of Mamba/SSMs in remote sensing representation learning.

The direct application of Mamba to ORSI-SOD exposes certain limitations. The blockwise modeling approach of Mamba lacks sufficient cross-boundary information flow, leading to blurred object contours and compromised representation of fine structural details [24], [48]. This limitation becomes especially severe when detecting small salient targets against cluttered backgrounds. Furthermore, existing VMamba-based remote sensing methods primarily focus on classification or dense prediction tasks [54], [55], where global semantic consistency is more important than fine-grained object localization. ORSI-SOD, however, demands both accurate global context understanding to handle multiscale heterogeneity and precise local feature discrimination to resolve low-contrast boundaries.

Motivated by these observations, we adopt VMamba [24] as the backbone and introduce correlation variance compensation tailored to ORSI-SOD. VMamba provides a hierarchical visual state-space encoder that naturally produces multiscale feature maps, making it well-suited to encoder–decoder saliency frameworks under large-scale variation. Compared with other vision–Mamba architectures, such as Vim [56] and the task-specific VM-UNet [57], VMamba is more backbone-oriented and can be integrated into our pipeline with minimal architectural modifications, allowing us to focus on enhancing feature discriminability and boundary preservation.

III. Proposed Method

A. Network Overview

The overall framework of MambaCVC is illustrated in Fig. 2. It adopts a classic encoder–decoder structure and integrates two key modules to enhance feature interaction: GC-VCM and LC-VCM. This design aims to improve the efficiency of global feature extraction and the detection of local details through principled correlation variance compensation.

Specifically, the encoder is based on the VMamba [24], with an input image size of $224 \times 224 \times 3$. To enhance spatial feature extraction and prevent small target loss through over-downsampling, we apply a minimal modification to the stem module of the original VMamba, adjusting the convolution kernel size and stride to increase the 2-D feature map resolution from $H/4 \times W/4$ to $H/2 \times W/2$, where H and W represent the height and width of the input image, respectively. The encoder consists of four visual state-space (VSS) blocks, denoted as *Enc-i* ($i \in \{1, 2, 3, 4\}$), with output feature sizes of $112 \times 112 \times 128$, $56 \times 56 \times 256$, $28 \times 28 \times 512$, and $14 \times 14 \times 1024$, respectively. Multiscale feature maps from the encoder differ in resolution and channel count, making direct fusion computationally complex and prone to mismatches. To effectively model their correlations while maintaining computational efficiency, we unified the number of channels in each layer to 32 through convolution operations.

Following the encoder, the GC-VCM module enhances and interacts with these global features through correlation variance compensation. This module captures additional global contextual information across different feature maps by explicitly modeling the statistical variations in interlayer correlations, enabling the model to consolidate long-range dependencies while suppressing homogeneous background patterns. The subsequent LC-VCM module focuses on refining local details through localized variance compensation, designed to enhance local features critical for small-scale object detection and boundary precision. This module captures

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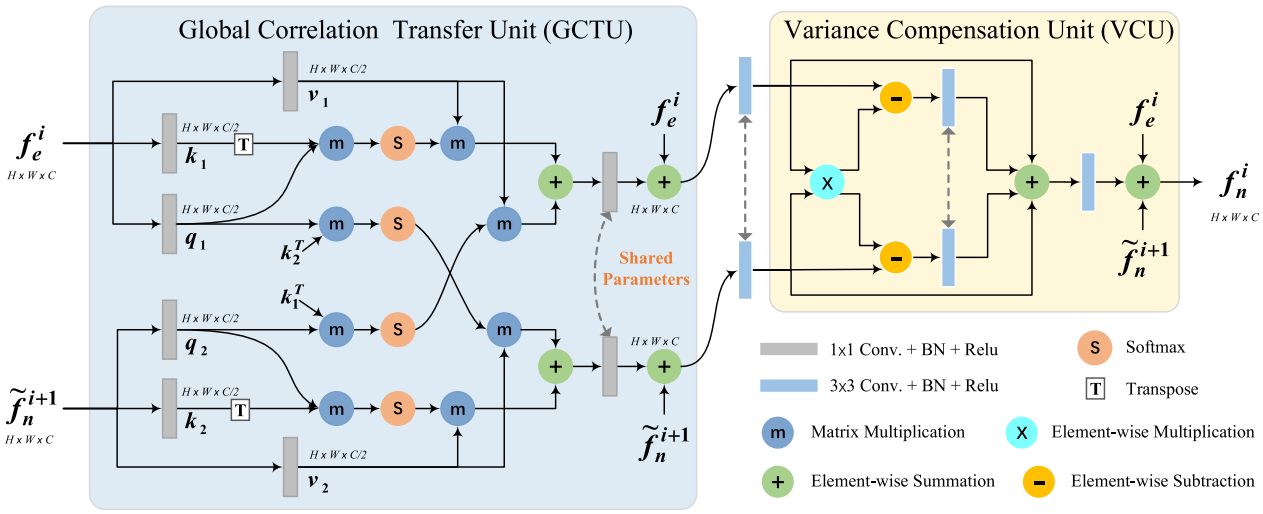


Fig. 3. Illustration of the GC-VCM.

and strengthens fine-grained spatial details by exploiting LC variance, ensuring accurate feature alignment across layers and improving sensitivity to subtle intensity differences.

Finally, the decoder gradually recovers the spatial resolution of the feature map through multiple decoding blocks, denoted as *Dec-i* ($i \in \{1, 2, 3, 4\}$), each containing three convolutional layers. By effectively combining global information extraction through variance-aware GC and local feature refinement through localized variance compensation, the entire network framework significantly improves the performance of ORSI-SOD in complex scenarios.

B. GC Variance Compensation Module

In ORSI-SOD tasks, features at different scales encode complementary semantic and structural information. Although VMamba enables multiscale feature extraction with long-range dependencies, the correlations between these feature levels often exhibit significant statistical variations that contain crucial discriminative information. From an information-theoretic perspective, *correlation variance represents the uncertainty in interlayer feature relationships*—regions with high correlation variance typically correspond to semantically important boundaries and object transitions, while low-variance regions often represent homogeneous backgrounds. Existing methods overlook this valuable statistical information, leading to suboptimal feature fusion and missed detection opportunities.

To exploit this insight, we propose the GC-VCM, which enhances interscale feature interaction through principled correlation variance compensation. GC-VCM consists of two synergistic components: the GCTU and the VCU. GCTU establishes global semantic alignment across different feature scales through cross-scale attention mechanisms, enabling the network to exploit long-range dependencies more effectively. VCU then models and compensates for correlation variance, ensuring that discriminative information encoded in interlayer correlation statistics is preserved and enhanced rather than suppressed during integration. This joint design improves

Algorithm 1 GC Transfer Unit

Input: Input feature maps: f_e^i, f_n^{i+1}
Output: Output feature maps: f_{gct}^i, f_{gct}^{i+1}

Step 1: Input Feature Alignment
  $\tilde{f}_n^{i+1} \leftarrow c_{1\times1}(Upsample(f_n^{i+1}))$
Step 2: Query, Key, and Value Generation
  $q_1 \leftarrow c_{1\times1}(f_e^i)$ $q_2 \leftarrow c_{1\times1}(\tilde{f}_n^{i+1})$
  $k_1 \leftarrow c_{1\times1}(f_e^i)$ $k_2 \leftarrow c_{1\times1}(\tilde{f}_n^{i+1})$
  $v_1 \leftarrow c_{1\times1}(f_e^i)$ $v_2 \leftarrow c_{1\times1}(\tilde{f}_n^{i+1})$
Step 3: Cross-scale Attention Calculation
  $s_{1,1} \leftarrow softmax(q_1 \cdot k_1^T)$ $s_{1,2} \leftarrow softmax(q_1 \cdot k_2^T)$
  $s_{2,1} \leftarrow softmax(q_2 \cdot k_1^T)$ $s_{2,2} \leftarrow softmax(q_2 \cdot k_2^T)$
Step 4: Global Feature Correlation Transfer
  $\hat{f}_e^i \leftarrow s_{1,1} \cdot v_1 + s_{2,1} \cdot v_1$
  $\hat{f}_n^{i+1} \leftarrow s_{1,2} \cdot v_2 + s_{2,2} \cdot v_2$
Step 5: Feature Fusion and Output
  Apply shared Parameters $c_{1\times1}$ for both outputs
  $f_{gct}^i \leftarrow c_{1\times1}(\hat{f}_e^i) + f_e^i$
  $f_{gct}^{i+1} \leftarrow c_{1\times1}(\hat{f}_n^{i+1}) + \tilde{f}_n^{i+1}$
Return f_{gct}^i, f_{gct}^{i+1}

global consistency in feature representation while maintaining sensitivity to subtle but important variations.

1) GC Transfer Unit: GCTU captures global dependencies between different feature layers through a cross-scale attention mechanism, enabling effective global information transfer across hierarchical representations. The detailed structure of the GCTU is shown in Fig. 3 (left).

The processing flow of GCTU is illustrated in Algorithm 1. First, the module receives two adjacent input feature layers, denoted as f_e^i and f_n^{i+1}, where f_e^i is the output feature of the encoder after channel unification, and f_n^i is the feature enhanced by the GC-VCM module. The index $i \in \{1, 2, 3\}$, and when $i = 3$, $f_n^{i+1} = f_e^4$. Since these feature layers originate from different scales, the feature map f_n^{i+1} undergoes upsampling and a 1×1 convolution operation followed by

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batch normalization (BN) and ReLU activation to align its spatial dimensions with the higher-level feature map f_e^i.

Next, the module generates corresponding queries (q), keys (k), and values (v) for both feature layers through 1×1 convolutions, mapping features from different scales into unified representational spaces. Cross-scale attention weights are computed through matrix multiplication between queries and keys, followed by softmax normalization to ensure proper probability distributions. These attention weights are then applied to the values to achieve cross-scale feature interaction, generating enhanced feature maps $\hat{f}_e^i$ and $\hat{f}_n^{i+1}$ that incorporate global information transfer across scales.

The enhanced features are fused with the original features through 1×1 convolution and residual connections to generate the final output feature maps f_{gct}^i and f_{gct}^{i+1}, ensuring both enhanced GC and preserved original semantic information.

2) Variance Compensation Unit: The VCU introduces a paradigm for feature fusion in ORSI-SOD by explicitly modeling and compensating for correlation variance between adjacent feature layers. Contemporary approaches predominantly focus on correlation enhancement through direct feature concatenation, attention-based weighting, or arithmetic operations, thereby overlooking the discriminative potential in interlayer correlation statistics [58], [59]. Meanwhile, lightweight feature interaction/selection has also been explored to enhance VMamba representations [60]. Such designs mainly emphasize interaction/selection to strengthen feature coupling, whereas our VCU explicitly models interlayer correlation variance and compensates it as a discriminative statistic to drive variance-compensation-driven local interaction and fusion.

The theoretical foundation derives from the classical interlayer correlation coefficient

$$\rho\left(\mathbf{F}_i, \mathbf{F}_{i+1}\right) = \frac{\operatorname{Cov}\left(\mathbf{F}_i, \mathbf{F}_{i+1}\right)}{\sqrt{\operatorname{Var}\left(\mathbf{F}_i\right)\ \operatorname{Var}\left(\mathbf{F}_{i+1}\right)}} \tag{1}$$

where $\operatorname{Cov}(\mathbf{F}_i, \mathbf{F}_{i+1})$ denotes the covariance between the two feature layers, and $\operatorname{Var}(\mathbf{F}_i)$ and $\operatorname{Var}(\mathbf{F}_{i+1})$ represent the variances of the respective layers. The key insight is that spatial variations in correlation strength across local neighborhoods encode essential discriminative information: regions with high correlation variance typically correspond to semantic boundaries with inconsistent interlayer relationships, while homogeneous areas exhibit low correlation variance with stable correspondences.

Our VCU implementation addresses these challenges through a learnable approximation that preserves the theoretical essence while ensuring efficiency and robustness. Given correlation-enhanced feature maps $\mathbf{f}_{gct}^i$ and $\mathbf{f}_{gct}^{i+1}$, the unit establishes a correlation baseline

$$f_{mul}^i = c_{3\times3}\left(f_{gct}^i\right) \otimes c_{3\times3}\left(f_{gct}^{i+1}\right) \tag{2}$$

where $c_{3\times3}$ denotes a 3×3 convolution with BN and ReLU activation, and $\otimes$ is elementwise multiplication. This serves as a learnable proxy to the covariance numerator in (1), identifying regions of strong coactivation. Subsequently, asymmetric differential responses isolate variance-sensitive patterns

$$f_{vcu}^i = c_{3\times3}\left(c_{3\times3}\left(f_{gct}^{i+1}\right) \ominus f_{mul}^i\right) \oplus c_{3\times3}\left(f_{gct}^i\right) \tag{3}$$

$$f_{vcu}^{i+1} = c_{3\times3}\left(c_{3\times3}\left(f_{gct}^i\right) \ominus f_{mul}^i\right) \oplus c_{3\times3}\left(f_{gct}^{i+1}\right) \tag{4}$$

where $\ominus$ and $\oplus$ denote elementwise subtraction and summation, respectively. The subtraction operations isolate deviations from the correlation baseline, capturing variance-analogous patterns corresponding to the variance terms in (1). Convolutional refinement transforms these patterns into robust discriminative features through adaptive compensation kernels. Final integration synthesizes compensated signals with original representations

$$\hat{f}_{vcu}^i = c_{3\times3}\left(f_{vcu}^i \oplus f_{vcu}^{i+1}\right) \tag{5}$$

$$f_n^i = \hat{f}_{vcu}^i \oplus f_e^i \oplus \tilde{f}_n^{i+1}. \tag{6}$$

The VCU architecture fundamentally differs from existing variance-processing methodologies through its dual-baseline compensation strategy. Traditional variance normalization suppresses variance for stability, while attention mechanisms exploit variance primarily for weighting. Our approach treats correlation variance as an intrinsic discriminative signal requiring explicit modeling and enhancement. Unlike correlation-maximization methods that emphasize interlayer consistency while sacrificing boundary-informative variations, VCU employs f_{mul}^i as a learnable reference baseline from which meaningful statistical deviations are extracted. This formulation preserves subtle yet informative activations beyond common response regions while enabling adaptive balancing between stable semantic correlations and variant boundary patterns. The convolutional processing implements learnable variance compensation kernels that adaptively transform deviation patterns according to local contexts, achieving noise robustness while maintaining sensitivity to subtle boundary transitions.

The VCU represents a computationally tractable approximation of the correlation-variance paradigm in (1). Through integrated modeling of correlation baselines and variance deviations, it systematically emphasizes high-variance boundary regions while modulating homogeneous correlation patterns. This combination of correlation theory with adaptive neural compensation establishes theoretical grounding while delivering practical advantages over conventional correlation-enhancement or variance-suppression techniques in complex ORSI-SOD applications.

C. Local Correlation Variance Compensation Module

While GC-VCM addresses GC variance, ORSI-SOD requires additional mechanisms to handle fine-grained local feature variations that are critical for accurate boundary delineation and small object detection. Local features exhibit their own correlation variance patterns that encode boundary information and subtle intensity differences characteristic of remote sensing imagery. The LC-VCM addresses this need through localized variance compensation mechanisms that preserve fine-grained discriminative information while enhancing local feature interactions.

The LC-VCM consists of the LC transfer unit (LCTU) and a localized VCU. The LCTU captures multiscale local dependencies through diversified convolutional processing, while the

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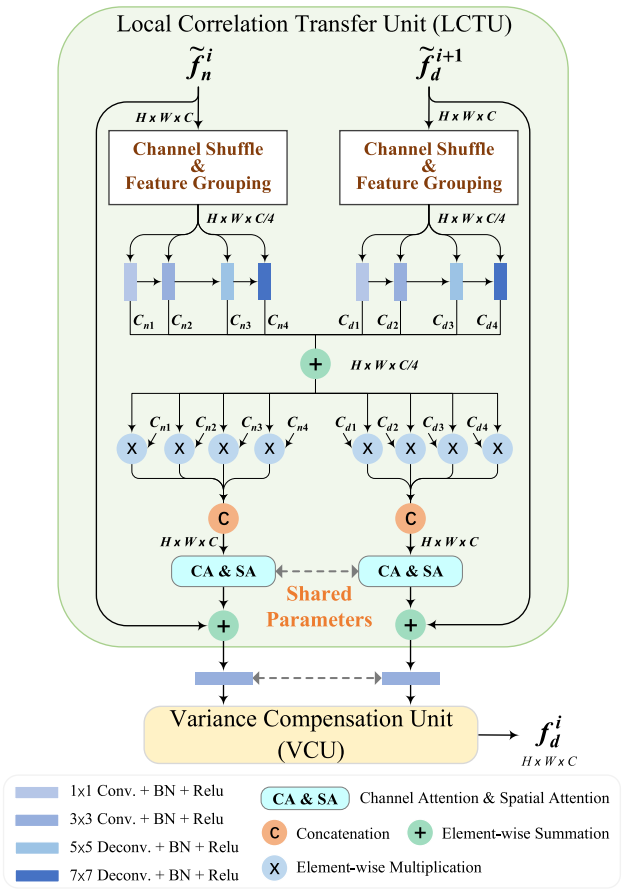


Fig. 4. Schematic of the LC-VCM.

VCU applies the same variance compensation principles at the local level to enhance boundary-sensitive features.

LCTU: The LCTU captures local dependencies through multiscale convolutions and cross-layer feature interactions, as detailed in Fig. 4. The processing flow follows Algorithm 2.

The module receives two adjacent input feature layers, denoted as f_n^i and f_d^{i+1}, where the index $i \in \{1, 2, 3\}$ represents the sequence of feature layers. Since the two feature layers come from different scales, the lower-layer feature map f_d^{i+1} first undergoes an upsampling operation, followed by a 1×1 convolution with BN and ReLU, to align its spatial dimensions with the higher-layer feature map f_n^i. Simultaneously, f_n^i is processed with a 1×1 convolution (BN+ReLU). The processed feature maps are denoted as $\tilde{f}_n^i$ and $\tilde{f}_d^{i+1}$.

Next, the module applies a channel shuffle operation on the processed feature maps to facilitate feature grouping. After shuffling, the feature maps $\tilde{f}_n^i$ and $\tilde{f}_d^{i+1}$ are split into four subfeature maps, denoted as f_{n1}–f_{n4} and f_{d1}–f_{d4}, respectively. Each of these subfeature maps then undergoes local convolutional processing with different kernel sizes: 1×1, 3×3, 5×5, and 7×7 convolutions. In particular, the 5×5 and 7×7 convolutions use dilation rates of 2 and 3, respectively, to enhance the spatial representation of local features.

In both branches, the subfeature maps f_{n1}–f_{n4} and f_{d1}–f_{d4} are processed through iterative convolutional layers. Each layer incorporates information from the preceding layer through addition, ensuring a gradual enhancement of local features.

Algorithm 2 LC Transfer Unit

Input: Input feature maps: f_n^i, f_d^{i+1}
Output: Output feature maps: f_{lct}^i, f_{lct}^{i+1}

Step 1: Initial Feature Processing
  $\tilde{f}_n^i \leftarrow c_{1\times1}(f_n^i)$ $\tilde{f}_d^{i+1} \leftarrow c_{1\times1}(Upsample(f_d^{i+1}))$
Step 2: Channel Shuffle and Split
  $f_{n1}, f_{n2}, f_{n3}, f_{n4} \leftarrow ChannelShuffle(\tilde{f}_n^i, 4)$
  $f_{d1}, f_{d2}, f_{d3}, f_{d4} \leftarrow ChannelShuffle(\tilde{f}_d^{i+1}, 4)$
Step 3: Local Convolutional Branch Processing
  For f_n^i **branch:**
    $C_{n1} \leftarrow c_{1\times1}(f_{n1})$ $C_{n2} \leftarrow c_{3\times3}(f_{n2} + C_{n1})$
    $C_{n3} \leftarrow c_{5\times5}(f_{n3} + C_{n2})$ $C_{n4} \leftarrow c_{7\times7}(f_{n4} + C_{n3})$
  For f_d^{i+1} **branch:**
    $C_{d1} \leftarrow c_{1\times1}(f_{d1})$ $C_{d2} \leftarrow c_{3\times3}(f_{d2} + C_{d1})$
    $C_{d3} \leftarrow c_{5\times5}(f_{d3} + C_{d2})$ $C_{d4} \leftarrow c_{7\times7}(f_{d4} + C_{d3})$
Step 4: Cross-Layer Feature Interaction
  Calculate shared feature sum:
  $f_{sum} = C_{n1} + C_{n2} + C_{n3} + C_{n4} + C_{d1} + C_{d2} + C_{d3} + C_{d4}$
  Apply multiplication for feature interaction:
  For f_n^i **branch:**
    $\hat{f}_{n1} \leftarrow C_{n1} * f_{sum}$ $\hat{f}_{n2} \leftarrow C_{n2} * f_{sum}$
    $\hat{f}_{n3} \leftarrow C_{n3} * f_{sum}$ $\hat{f}_{n4} \leftarrow C_{n4} * f_{sum}$
  For f_d^{i+1} **branch:**
    $\hat{f}_{d1} \leftarrow C_{d1} * f_{sum}$ $\hat{f}_{d2} \leftarrow C_{d2} * f_{sum}$
    $\hat{f}_{d3} \leftarrow C_{d3} * f_{sum}$ $\hat{f}_{d4} \leftarrow C_{d4} * f_{sum}$
Step 5: Shared Channel and Spatial Attention
  Apply shared CA and SA to both branches:
  $\hat{f}_{lct}^i = SA(CA(\text{Concat}(\hat{f}_{n1}, \hat{f}_{n2}, \hat{f}_{n3}, \hat{f}_{n4})))$
  $\hat{f}_{lct}^{i+1} = SA(CA(\text{Concat}(\hat{f}_{d1}, \hat{f}_{d2}, \hat{f}_{d3}, \hat{f}_{d4})))$
Step 6: Residual Connection
  $f_{lct}^i = \hat{f}_{lct}^i + \tilde{f}_n^i$ $f_{lct}^{i+1} = \hat{f}_{lct}^{i+1} + \tilde{f}_d^{i+1}$
Return f_{lct}^i, f_{lct}^{i+1}

The outputs from these operations are denoted as C_{n1}–C_{n4} and C_{d1}–C_{d4}.

The next step is cross-layer feature interaction, where the module calculates a shared feature sum, f_{sum}, by summing all the convolution results from both branches. This feature sum is then applied through elementwise multiplication to each subfeature map in both branches, resulting in enhanced interaction across layers. The interaction-enhanced features for f_n^i and f_d^{i+1} branches are denoted as $\hat{f}_{n1}$–$\hat{f}_{n4}$ and $\hat{f}_{d1}$–$\hat{f}_{d4}$, respectively.

Following the interaction, the module applies shared channel attention (CA) and spatial attention (SA) modules to the concatenated interaction features from each branch. This produces the enhanced feature maps $\hat{f}_{lct}^i$ and $\hat{f}_{lct}^{i+1}$.

Finally, the enhanced features $\hat{f}_{lct}^i$ and $\hat{f}_{lct}^{i+1}$ are added back to the initial processed features $\tilde{f}_n^i$ and $\tilde{f}_d^{i+1}$ through residual connections, yielding the final output feature maps f_{lct}^i and f_{lct}^{i+1}. This design effectively integrates multiscale local features, enhancing the feature representation of ORSI-SOD.

D. Training Strategy

1) Loss Function: As illustrated in Fig. 1, we employ the widely-used deep supervision during the training phase,

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monitoring intermediate salient maps of different scales to compel the feature learning of salient regions of various scales. Classic pixel-level *binary cross-entropy* (BCE) loss and image-level *intersection over union* (IoU) loss are employed in our loss function for network training. The loss for each decoder block of MambaCVC is formulated as follows:

$$\mathcal{L}_{sal}^{t}\left(s^{t}, g\right) = \mathcal{L}_{bce}\left(s^{t}, g\right) + \mathcal{L}_{iou}\left(s^{t}, g\right). \tag{7}$$

Here, g represents the ground truth (GT), and s^t is the salient map of the tth decoded block of MambaCVC, where $t \in \{1, 2, 3, 4\}$. $\mathcal{L}_{bce}(\cdot)$ and $\mathcal{L}_{iou}(\cdot)$ denote the BCE loss and IoU loss, respectively.

2) Mixup Data Augmentation Strategy: Mixup [61] is a data augmentation strategy that generates virtual training samples by linearly interpolating two input–label pairs. Given two samples (s_i, g_i) and (s_j, g_j), we construct a mixed input as

$$\tilde{s} = \lambda s_i + (1 - \lambda)\, s_j, \quad \lambda \in [0, 1] \tag{8}$$

where λ is usually drawn from a Beta distribution. In our implementation, we fix $\lambda = 0.4$ as a moderate mixing ratio to balance regularization and structural preservation, which leads to stable training and consistent improvements in our setting.

We emphasize that Mixup is applied only at the input level (image–mask pairs). That is, we do not mix intermediate feature maps. Instead, for the mixed input $\tilde{s}$, the supervision is implemented by linearly combining the losses with respect to the two original GT masks. Concretely, with deep supervision across decoder stages, the tth decoder output on $\tilde{s}$ is denoted as $\tilde{s}^t$, and the total loss is defined as

$$\tilde{\mathcal{L}}_{\text{total}} = \sum_{t=1}^{4} \left(\lambda\, \mathcal{L}_{\text{sal}}^{t}\left(\tilde{s}^{t}, g_i\right) + (1 - \lambda)\, \mathcal{L}_{\text{sal}}^{t}\left(\tilde{s}^{t}, g_j\right)\right) \tag{9}$$

where $\mathcal{L}_{\text{sal}}^{t}$ is the saliency loss applied at the tth decoder stage. In this way, Mixup interacts with deep supervision by using the same mixing coefficient to weight the supervision signals at each decoder stage, ensuring consistent training across multilevel predictions.

IV. Experiment

A. Experimental Setup

1) Datasets: We performed comprehensive training and testing for the proposed MambaCVC using three publicly available datasets designed for the ORSI-SOD task: the ORSSD dataset [25], the EORSSD dataset [26], and the ORSI-4199 dataset [28]. Specifically, the ORSSD dataset contains 800 images, of which 600 are for training and 200 are for testing, and is the first basic benchmark proposed in this field. The EORSSD dataset extends the ORSSD to 2000 images, of which 1400 are used for training and 600 are used for testing, thus introducing more diverse and complex scenarios to improve the generalization ability of the model. Finally, the ORSI-4199 dataset contains 4199 images, of which 2000 are used for training and 2199 for testing, representing more complex scenarios reflecting real-world conditions.

2) Implementation Details: We augmented the training set by rotating images at angles of 90°, 180°, and 270°, as well as performing mirror reflections followed by rotations at angles of 90°, 180°, and 270°. In addition, during the training phase, each image was resized to a fixed size of 224 × 224.

Experiments with MambaCVC were conducted on the PyTorch platform using NVIDIA Tesla V100 (32 GB memory). The encoder was initialized with the pretrained parameters of VMamba-Base. In addition, MambaCVC uses the AdamW [62] optimization strategy; the batch size is 16; the initial learning rate is $1e^{-4}$; the minimum learning rate is $1e^{-6}$; the learning rate scheduling method is CosineAnnealingLR with warm-up epochs [63]; and the maximum epoch is set to 100.

3) Evaluation Metrics: A more comprehensive quantitative comparison is performed on three publicly available datasets for different ORSI-SOD models, and we use widely used evaluation metrics to evaluate MambaCVC and other comparative methods. These metrics include: S-measure ($S_\alpha, \alpha = 0.5$), F-measure ($F_\beta^{max}, \beta^2 = 0.3$), E-measure ($E_\xi^{max}$), mean absolute error (MAE, $\mathcal{M}$), precision–recall (PR) curve, parameters (#Param), and Flops. These metrics are utilized to assess the performance of ORSI-SOD models in various aspects.

B. Comparison With State-of-the-Art Methods

In this section, we perform a quantitative comparison on three publicly available datasets (ORSSD, EORSSD, and ORSI-4199). The detection results of all compared methods are based on publicly available pretrained models or provided by the authors. To ensure fairness in the evaluation, all methods are trained and tested using a consistent strategy, including the same data augmentation method, optimizer configuration, and evaluation metrics.

1) Quantitative Comparisons: We comprehensively analyze the proposed MambaCVC with 16 state-of-the-art ORSI-SOD methods on the ORSSD and EORSSD datasets. This includes four CNN-based methods (UG2L [64], ADSTNet [65], WeightNet [35], and RAGRNet [36]), four Transformer-based methods (GeleNet [17], BMGT [41], TLCKD [18], and PRNet [66]), four hybrid CNN–Transformer methods (HFANet [43], GSANet [42], DCPNet [44], and HFCNet [45]), and four lightweight CNN-based methods (SeaNet [38], MEANet [67], SAFINet [59], and RAMENet [58]). Our method demonstrates significant advantages on both the ORSSD and EORSSD datasets, as shown in Table I. In the ORSSD dataset, MambaCVC achieves the highest scores in the F_β^{max} and E_ξ^{max} metrics, reaching 0.9307 and 0.9890, respectively, ranking first among all methods. In addition, MambaCVC attains a close-to-optimal score in S_α, only slightly below HFCNet by 0.0028 (0.9543 versus 0.9515), indicating a very similar level of accuracy. Furthermore, MambaCVC maintains a low $\mathcal{M}$ of 0.0080, with only a small difference from the best score of 0.0066. In the EORSSD dataset, MambaCVC achieves the highest F_β^{max} score of 0.8982 and the second-best score of 0.9437 in S_α, showing near-optimal performance. Notably, in both E_ξ^{max} and $\mathcal{M}$ metrics, our method attains impressive scores of 0.9813 and 0.0056, with minimal difference from the best-performing methods.

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TABLE I

QUANTITATIVE COMPARISON OF MAMBACVC WITH ADVANCED ORSI-SOD METHODS ON THE ORSSD AND EORSSD DATASETS. THE METHODS COMPARED INCLUDE THE FOLLOWING TYPES: CNN-BASED METHODS (CR), TRANSFORMER-BASED METHODS (TR), HYBRID CNN–TRANSFORMER METHODS (HR), LIGHTWEIGHT CNN-BASED METHODS (LR), AND MAMBA-BASED METHODS (MR). ↑ MEANS BIGGER IS BETTER AND ↓ MEANS SMALLER IS BETTER. THE TOP THREE RESULTS ARE DENOTED IN RED, BLUE, AND GREEN, RESPECTIVELY

Methods	Type	Backbone	#Param $(M)\downarrow$	FLOPs $(G)\downarrow$	ORSSD Dataset				EORSSD Dataset			
					$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$
UG2L [64]	CR	ResNet	57.2	102.5	.9405	.9130	.9784	.0098	.9376	.8856	.9723	.0069
ADSTNet [65]	CR	Res2Net	60.1	27.7	.9379	.9124	.9807	.0086	.9311	.8804	.9769	.0065
WeightNet [35]	CR	ResNet	120.7	66.1	.9478	.9200	.9807	.0082	.9380	.8885	.9782	.0060
RAGRNet [36]	CR	Res2Net	35.6	17.8	.9507	.9242	.9861	.0066	.9361	.8852	.9785	.0057
GeleNet [17]	TR	PVT	25.5	11.7	.9469	.9254	.9860	.0079	.9376	.8923	.9828	.0064
BMGT [41]	TR	Segformer	3.8	22.7	.9368	.9105	.9781	.0092	.9292	.8829	.9740	.0058
TLCKD [18]	TR	ResT-Large	50	32	.9421	.9114	.9794	.0082	.9350	.8843	.9788	.0056
PRNet [66]	TR	PVT	20.8	8.47	.9459	.9177	.9848	.0075	.9276	.8684	.9784	.0054
HFANet [43]	HR	Hybrid	60.5	68.3	.9399	.9112	.9770	.0092	.9380	.8876	.9740	.0070
GSANet [42]	HR	Hybrid	49.5	11.4	.9491	.9253	.9864	.0070	.9391	.8937	.9836	.0053
DCPNet [44]	HR	Hybrid	99.3	20.5	.9498	.9251	.9855	.0073	.9408	.8936	.9817	.0053
HFCNet [45]	HR	Hybrid	140.8	120.4	.9543	.9269	.9884	.0070	.9449	.8951	.9842	.0049
SeaNet [38]	LR	MobileNet	2.76	1.74	.9260	.8942	.9767	.0105	.9208	.8649	.9710	.0073
MEANet [67]	LR	MobileNet	3.27	9.62	.9340	.9033	.9768	.0098	.9282	.8766	.9708	.0070
SAFINet [59]	LR	MobileNet	3.12	7.63	.9401	.9106	.9786	.0086	.9267	.8799	.9732	.0065
RAMENet [58]	LR	MobileViT	5.18	8.72	.9489	.9229	.9862	.0072	.9419	.8971	.9822	.0050
MambaCVC	**MR**	**VMamba**	**7.26**	**6.25**	.9515	.9307	.9890	.0080	.9437	.8982	.9813	.0056
Ours-Res	**CR**	**ResNet**	**25.04**	**17.67**	.9462	.9221	.9828	.0088	.9371	.8910	.9772	.0068
Ours-PVT	**TR**	**PVT**	**25.46**	**11.35**	.9495	.9240	.9865	.0073	.9410	.8932	.9801	.0055
Ours-Hyb	**HR**	**Hybrid**	**49.87**	**20.90**	.9479	.9232	.9862	.0072	.9402	.8930	.9821	.0055
Ours-Mob	**LR**	**MobileNet**	**2.77**	**3.64**	.9388	.9094	.9752	.0092	.9295	.8803	.9746	.0072

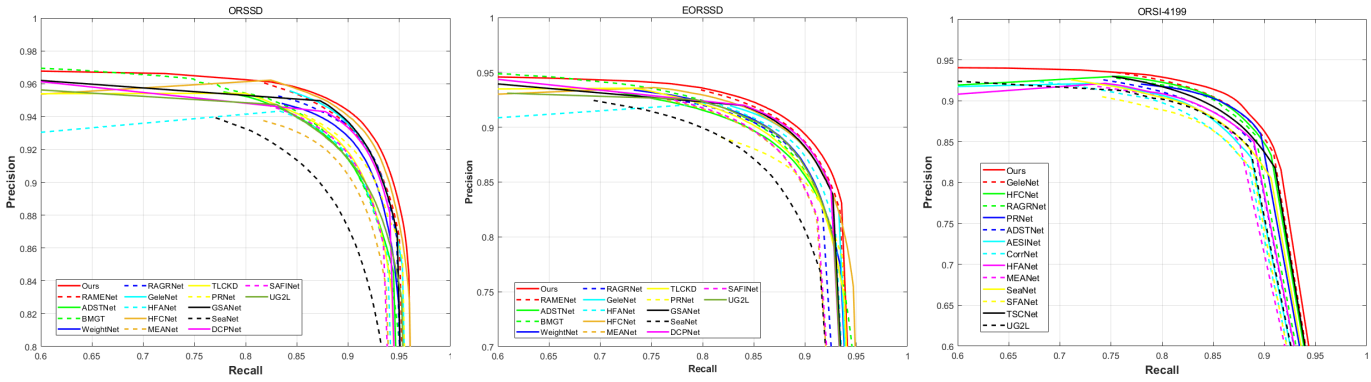


Fig. 5. Quantitative comparison of PR curves for the three datasets. We show the top five in different colors, and the other compared methods in gray.

On the ORSI-4199 dataset, we quantitatively compared MambaCVC with 13 state-of-the-art ORSI-SOD methods (AESINet [33], TSCNet [68], UG2L [64], SFANet [69], ADSTNet [65], RAGRNet [36], GeleNet [17], PRNet [66], HFANet [43], HFCNet [45], CorrNet [37], SeaNet [38], and MEANet [67]), whose saliency maps were available on the ORSI-4199 dataset. The proposed MambaCVC demonstrates outstanding performance across all evaluation metrics, as shown in Table II. In terms of S_α, MambaCVC achieves the highest score of 0.8928, underscoring its superior capability in capturing the structural details of salient objects. For F_β^{max}, MambaCVC again leads with a top score of 0.8904, indicating highly balanced detection accuracy. In the E_ξ^{max} metric, MambaCVC excels with a score of 0.9565, showcasing its strength in precise pixel correspondence. Finally, MambaCVC maintains a low $\mathcal{M}$ of 0.0269, ranking closely behind the best method by a minimal margin.

In addition, we plot the PR curves of MambaCVC and all compared methods on the three datasets. The closer the PR curve is to the coordinate (1, 1), the better the performance of the network. As can be seen in Fig. 5, the PR curve of our method (i.e., the red solid line) is closest to the top-right corner on all three datasets.

To comprehensively assess the effectiveness and generality of our framework, we further implemented four variants with different backbone architectures, namely ResNet50 (Ours-Res), Pyramid Vision Transformer (Ours-PVT), MobileNet (Ours-Mob), and a hybrid CNN–Transformer backbone

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TABLE II

QUANTITATIVE COMPARISON OF MAMBACVC WITH ADVANCED ORSI-SOD METHODS ON THE ORSI-4199 DATASETS. THE METHODS COMPARED INCLUDE THE FOLLOWING TYPES: CNN-BASED METHODS (CR), TRANSFORMER-BASED METHODS (TR), HYBRID CNN–TRANSFORMER METHODS (HR), LIGHTWEIGHT CNN-BASED METHODS (LR), AND MAMBA-BASED METHODS (MR). ↑ MEANS BIGGER IS BETTER AND ↓ MEANS SMALLER IS BETTER. THE TOP THREE RESULTS ARE DENOTED IN RED, BLUE, AND GREEN, RESPECTIVELY

Methods	Type	ORSI-4199 Dataset			
		$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
AESINet [33]	CR	.8702	.8670	.9403	.0325
TSCNet [68]	CR	.8785	.8772	.9487	.0295
UG2L [64]	CR	.8731	.8619	.9362	.0318
SFANet [69]	CR	.8762	.8710	.9447	.0292
ADSTNet [65]	CR	.8710	.8698	.9433	.0318
RAGRNet [36]	CR	.8811	.8811	.9492	.0284
GeleNet [17]	TR	.8862	.8842	**.9544**	**.0264**
PRNet [66]	TR	.8873	.8819	.9527	.0272
HFANet [44]	HR	.8767	.8700	.9431	.0314
HFCNet [43]	HR	.8827	.8831	.9525	.0281
CorrNet [37]	LR	.8623	.8560	.9330	.0366
SeaNet [38]	LR	.8722	.8653	.9426	.0308
MEANet [67]	LR	.8708	.8662	.9417	.0312
MambaCVC	**MR**	**.8928**	**.8904**	**.9565**	.0269
Ours-Res	**CR**	.8820	.8749	.9459	.0293
Ours-PVT	**TR**	**.8905**	**.8895**	**.9559**	**.0253**
Ours-Hyb	**HR**	**.8879**	**.8872**	.9534	**.0261**
Ours-Mob	**LR**	.8739	.8687	.9438	.0319

(Ours-Hyb), while consistently retaining the dual-correlation VCMs (GC-VCM and LC-VCM). It should be noted that existing methods adopt diverse backbone networks, and a strictly backbone-agnostic comparison is inherently infeasible, as many fusion-based approaches are specifically tailored to leverage the characteristics of their chosen backbones. Despite this limitation, the results in Table I clearly demonstrate that our modules consistently provide notable performance improvements across all tested backbones. In particular, the VMamba-based architecture achieves the most favorable balance between accuracy and computational efficiency, highlighting its suitability as the default backbone in our framework.

2) Attribute Analysis on ORSI-4199: To further validate the robustness and generalization capability of the proposed method under diverse and complex scenarios, we conducted a detailed attribute-based evaluation on the ORSI-4199 dataset, which includes nine challenging scene categories. Table III presents the S-measure scores of MambaCVC compared with seven state-of-the-art ORSI-SOD models. As shown in the table, MambaCVC achieves the highest score across all nine attributes, demonstrating its consistent superiority and strong adaptability under various scene conditions. This result clearly verifies the effectiveness of the proposed correlation variance compensation mechanism in enhancing feature discriminability and maintaining robustness across different contexts.

Specifically, MambaCVC shows remarkable performance in scenarios involving big salient objects (BSO), complex

TABLE III

ATTRIBUTE-BASED PERFORMANCE COMPARISON RESULTS ON ORSI-4199 DATASET. S-MEASURE SCORES ARE GIVEN FOR SPECIFIC ATTRIBUTES. THE TOP TWO RESULTS ARE DENOTED IN RED AND BLUE, RESPECTIVELY

Attributes / Methods	BSO	CS	CSO	ISO	LCS	MSO	NSO	OC	SSO
ADSTNet [65]	.9132	.8968	.8901	.8640	.8443	.8397	.8878	.8487	.8350
RAGRNet [36]	.9190	.9065	.8949	.8736	**.8541**	.8627	.8978	.8558	.8535
GeleNet [17]	.9227	.9123	**.9087**	.8738	.8499	.8625	**.9035**	.8635	.8565
PRNet [66]	**.9238**	**.9124**	.9073	**.8746**	.8519	**.8710**	.9024	**.8734**	**.8649**
CorrNet [37]	.8903	.8858	.8641	.8401	.8331	.8495	.8768	.8582	.8496
SeaNet [38]	.9186	.8982	.8994	.8696	.8433	.8442	.8876	.8505	.8449
MEANet [67]	.9159	.8955	.8967	.8681	.8403	.8579	.8748	.8543	.8523
MambaCVC	**.9286**	**.9217**	**.9125**	**.8894**	**.8642**	**.8730**	**.9148**	**.8748**	**.8671**

scenes (CS), complex salient objects (CSO), and incomplete salient objects (ISO), owing to the GC-VCM, which effectively models cross-scale global dependencies while suppressing redundant background responses. In low-contrast scenes (LCS) and multiple salient object (MSO) scenarios, the model maintains high detection precision through variance-driven feature enhancement, effectively balancing accuracy and stability. Meanwhile, in narrow salient object (NSO), off-center (OC), and small salient object (SSO) conditions, the LC-VCM significantly improves boundary localization and fine-grained structural preservation. Overall, MambaCVC achieves the best performance across all nine challenging attributes, confirming the comprehensive advantages and strong generalization capability of the proposed dual-correlation variance compensation framework for complex ORSI-SOD tasks.

3) Computational Complexity Comparison: Network parameters (#Param) and FLOPs are used to measure the computational complexity of the network. The results of all the compared methods are obtained by testing existing public code or from public papers, as shown in Table I. We depict the performance (Max F-measure) and computational efficiency (parameters and FLOPs) of MambaCVC on the ORSSD dataset compared to state-of-the-art methods, as shown in Fig. 6. In this comparison, MambaCVC (marked with a red star) stands out by achieving a high Max F-measure (around 0.93) with only 6.25G FLOPs and approximately 7.26M parameters. This demonstrates that MambaCVC maintains exceptional performance while keeping both the parameter count and FLOPs significantly lower than most other methods, highlighting its efficiency in computational complexity.

Comparing MambaCVC with other methods of varying complexities further illustrates its optimal balance. For instance, HFCNet, which also performs well, has a much higher parameter count (140.8M) and 120.4G FLOPs, indicating a significantly greater computational cost for similar performance. In addition, methods like UG2L and WeightNet require considerably more parameters and FLOPs, yet they do not surpass MambaCVC's performance. In contrast, several methods with smaller parameter counts highlight the trade-offs more clearly. SeaNet has very low parameters (2.76M) and FLOPs (1.74G), but its Max F-measure remains below 0.90, indicating a substantial performance loss for lightweight design. MEANet (3.27M) and SAFINet (3.12M) also maintain

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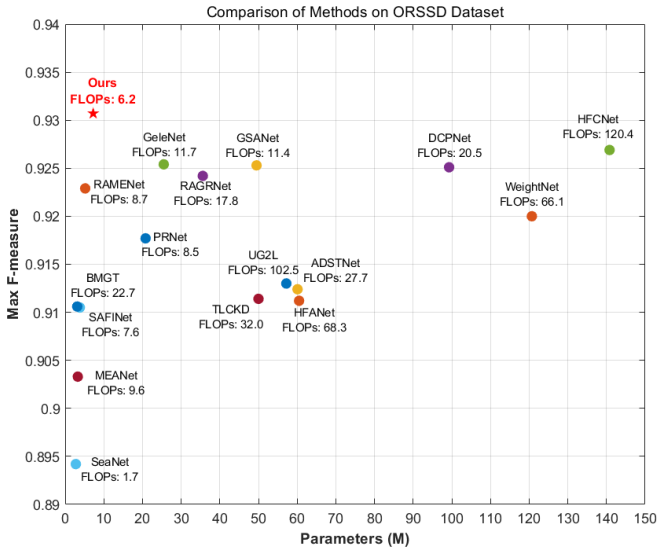


Fig. 6. Comparison of MambaCVC's performance (max F-measure) and computational efficiency (parameters and FLOPs) against state-of-the-art methods on the ORSSD dataset.

TABLE IV

REAL-TIME PERFORMANCE (FPS) COMPARISON OF MAMBACVC AND ITS VARIANTS. HARDWARE: GPU = NVIDIA TESLA V100 (32 GB) AND CPU = INTEL XEON SILVER 4110 @ 2.10 GHZ

Platform	MambaCVC	Ours-PVT	Ours-Mob
GPU FPS	19	28	34
CPU FPS	2	4	7

relatively low complexities, yet their Max F-measure scores are only around 0.91. Even BMGT (3.8M parameters and 22.7G FLOPs) reaches about 0.91 Max F-measure, but it likewise cannot match MambaCVC's performance despite its lightweight structure. RAMENet, with 5.18M parameters and 8.72G FLOPs, achieves higher Max F-measure values than these lightweight baselines, but still falls short of MambaCVC.

In summary, while several lightweight methods achieve low parameters and FLOPs, they fail to match the high detection accuracy of MambaCVC. By carefully balancing efficiency and complexity, MambaCVC surpasses both large, complex models and other lightweight approaches in the overall accuracy–efficiency tradeoff, demonstrating superior effectiveness under comprehensive evaluation.

4) Real-Time Performance Analysis: We further evaluated the frame rate of the proposed MambaCVC and its variants on both GPU and CPU platforms, with the results summarized in Table IV. On GPU, MambaCVC achieves 19 frames/s, while the variants Ours-PVT and lightweight Ours-Mob reach 28 and 34 frames/s, respectively. On CPU, MambaCVC runs at 2 frames/s, whereas Ours-Mob achieves 7 frames/s. These results indicate that the lightweight variants exhibit stronger real-time capability compared to the main model.

It is important to note that the primary goal of MambaCVC is to achieve a favorable balance between accuracy and complexity through dual-correlation variance compensation. With only 7.26M parameters and 6.25G FLOPs, MambaCVC significantly reduces theoretical computational complexity and achieves superior detection accuracy compared with existing methods. However, its real-time advantage has not been fully released. This discrepancy is largely related to the maturity of different backbones. PVT and MobileNet are well-established architectures that have been extensively optimized in mainstream deep learning frameworks and hardware libraries, often benefiting from advanced acceleration techniques such as FlashAttention [70] and optimized convolution kernels. As a result, they exhibit higher FPS in practice. In contrast, Mamba, as a recently introduced SSM, currently lacks such mature operator-level optimization. Although MambaCVC already demonstrates low theoretical complexity, its potential speed has not yet been fully exploited.

MambaCVC represents the optimal solution in terms of accuracy–complexity tradeoff, while its lightweight variants provide additional flexibility for resource-constrained scenarios. With future advances in hardware-aware optimization for Mamba modules, we expect that its lightweight design will be further translated into improved real-time performance.

5) Visual Comparison: In Fig. 7, we provide a visual comparison of MambaCVC with several representative state-of-the-art ORSI-SOD methods across challenging remote sensing scenarios. Each row in Fig. 7 highlights a distinct difficulty, allowing us to demonstrate the advantages of our approach.

In the first row, which depicts scenes containing multiple small targets, MambaCVC effectively captures each target with sharp boundaries and clear separation, whereas other methods often miss or merge small objects. The second row focuses on occluded targets. MambaCVC achieves complete detection of the entire salient objects without mistakenly including occlusion regions, a limitation observed in some comparative methods. The third row presents irregularly shaped targets. Our method excels in preserving their intricate shapes and contours, while other methods tend to produce incomplete or distorted object boundaries. The fourth row illustrates elongated targets. MambaCVC successfully delineates its full extent and structural integrity, showing strong adaptability to stretched or narrow forms. The fifth row highlights multiscale targets, where MambaCVC robustly captures both small and large objects with balanced clarity, unlike other approaches that often overemphasize larger targets or fail to detect the smaller ones. Finally, the sixth row depicts complex background scenarios. In these cases, MambaCVC isolates salient objects with minimal interference from cluttered surroundings, whereas alternative methods frequently confuse objects with the background. This visual comparison underscores the robustness and effectiveness of our method in capturing salient details even under challenging remote sensing conditions.

C. Ablation Studies

We conducted a comprehensive ablation study on the ORSSD dataset to evaluate the impact of each module in MambaCVC and the specific components within each module. All ablation studies were performed on MambaCVC with the modified VMamba backbone, using the same parameter settings and dataset partitioning as in Section IV-A.

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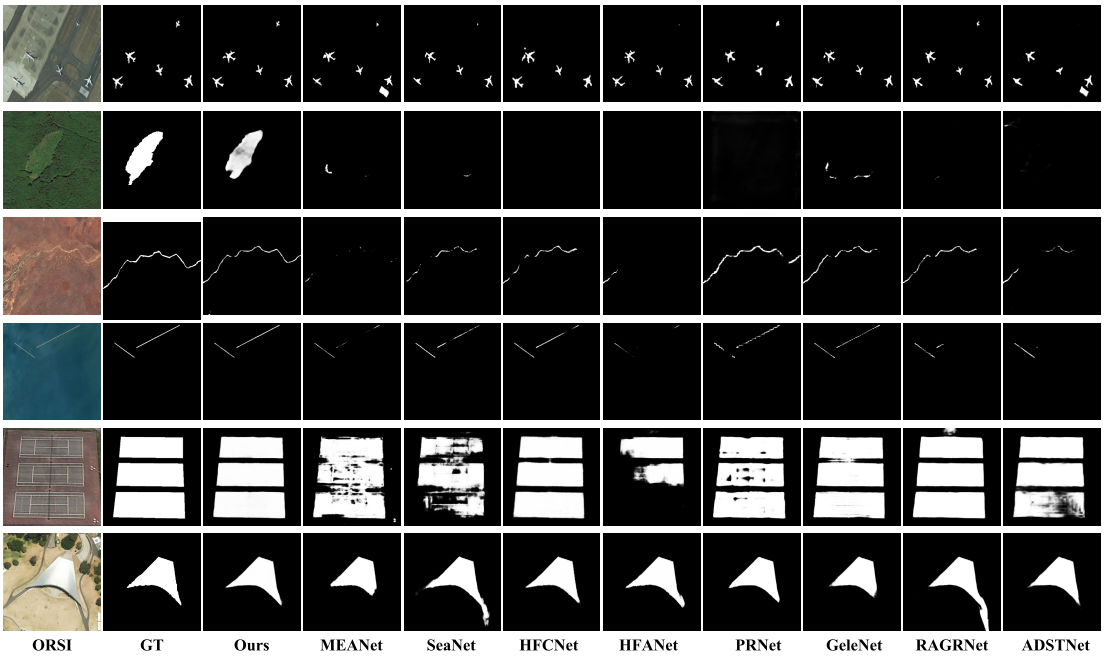


Fig. 7. Visual comparison of MambaCVC with representative state-of-the-art ORSI-SOD methods. Please zoom in for the best view.

TABLE V

QUANTITATIVE ABLATION ANALYSIS OF THE INDIVIDUAL CONTRIBUTIONS OF EACH MODULE IN MAMBACVC. THE BEST ONE IN EACH COLUMN IS IN BOLD

No.	Baseline	GC-VCM	LC-VCM	#Param $(M)\downarrow$	FLOPs $(G)\downarrow$	ORSSD Dataset			
						$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$
1	✓			7.12	1.34	.9147	.8976	.9618	.0217
2	✓*			7.05	4.40	.9355	.9141	.9713	.0156
3	✓*	✓		7.16	5.34	.9462	.9220	.9811	.0120
4	✓*		✓	7.15	5.31	.9475	.9256	.9853	.0116
5	✓*	✓	✓	7.26	6.25	**.9515**	**.9307**	**.9890**	**.0080**

✓*: Using the modified stem module of the original VMamba-Base

1) Individual Contribution of Each Module: In this section, we conducted extensive experiments to evaluate the effectiveness of key components within MambaCVC through an ablation study of various model configurations. These configurations include: 1) the "Baseline" model, consisting of only the encoder–decoder structure as an initial performance benchmark; 2) the "Modified Baseline" model, which enhances the baseline by incorporating a modified stem module from the original VMamba to optimize model structure; 3) "Modified Baseline + GC-VCM" model, which adds the GC-VCM to the Modified Baseline, aiming to enhance global feature representation; 4) "Modified Baseline + LC-VCM" model, which adds the LC-VCM to the Modified Baseline, improving the model's ability to capture local features and details; and 5) the MambaCVC Model, which is the complete model that includes the Modified Baseline, GC-VCM, and LC-VCM, integrating all components to achieve the best balance between performance and complexity.

Table V provides quantitative results of these configurations on the ORSSD dataset. The Baseline model has 7.12M parameters and 1.34G FLOPs, achieving $S_\alpha = 0.9147$, $F_\beta^{max} = 0.8976$, $E_\xi^{max} = 0.9618$, and $\mathcal{M} = 0.0217$ as a foundational benchmark. Adding the modified stem module in the Modified Baseline slightly reduces parameters to 7.05M but increases FLOPs to 4.40G, yielding improved performance with $S_\alpha = 0.9355$, $F_\beta^{max} = 0.9141$, $E_\xi^{max} = 0.9713$, and $\mathcal{M} = 0.0156$. Incorporating GC-VCM into the Modified Baseline increases parameters to 7.16M and FLOPs to 5.34G, further enhancing performance to $S_\alpha = 0.9462$, $F_\beta^{max} = 0.9220$, $E_\xi^{max} = 0.9811$, and $\mathcal{M} = 0.0120$, highlighting GC-VCM's role in global feature enhancement. Similarly, adding LC-VCM to the Modified Baseline results in 7.15M parameters and 5.31G FLOPs, achieving $S_\alpha = 0.9475$, $F_\beta^{max} = 0.9256$, $E_\xi^{max} = 0.9853$, and $\mathcal{M} = 0.0116$, demonstrating LC-VCM's effectiveness in capturing local details. Finally, the Full Model combines the Modified Baseline, GC-VCM, and LC-VCM, resulting in 7.26M parameters and 6.25G FLOPs. It achieves the best performance with $S_\alpha = 0.9515$, $F_\beta^{max} = 0.9307$, $E_\xi^{max} = 0.9890$, and $\mathcal{M} = 0.0080$, confirming the benefit of integrating all modules for optimal performance and efficiency. Overall, this ablation study demonstrates the significant contribution of GC-VCM and LC-VCM, which, when combined, enable substantial performance gains with minimal computational cost.

2) Effectiveness of Each Component in GC-VCM: The GC-VCM comprises two key components: GCTU and VCU. To evaluate the specific impact of each component on both accuracy and efficiency, we design two ablation variants by individually removing GCTU and VCU, resulting in the "$w/o\ \ GCTU$" and "$w/o\ \ VCU$" configurations. In addition to the saliency metrics, we further report the model complexity in terms of Params and FLOPs, so as to quantify the computational overhead introduced by each component.

Table VI summarizes the results. After removing GCTU, the model achieves $S_\alpha = 0.9488$, $F_\beta^{max} = 0.9229$, and $E_\xi^{max} = 0.9836$, while $\mathcal{M}$ increases to 0.0100, with Params/FLOPs

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TABLE VI

QUANTITATIVE ABLATION ANALYSIS OF THE EFFECTIVENESS OF EACH COMPONENT IN THE PROPOSED MODULES. THE BEST ONE IN EACH COLUMN IS IN BOLD

Modules	Components	#Param (M) ↓	FLOPs (G) ↓	ORSSD Dataset			
				S_α ↑	F_β^{max} ↑	E_ξ^{max} ↑	$\mathcal{M}$ ↓
GC-VCM	*w/o GCTU*	7.22	5.79	.9488	.9229	.9836	.0100
	w/o VCU	7.21	5.78	.9491	.9222	.9845	.0088
LC-VCM	*w/o LCTU*	7.22	5.83	.9481	.9236	.9843	.0095
	w/o VCU	7.21	5.78	.9486	.9244	.9840	.0092
	Groups = 1	7.26	6.22	.9496	.9298	.9865	.0084
	Groups = 2	7.27	6.27	.9505	.9301	.9876	.0081
	Groups = 4	7.26	6.25	**.9515**	**.9307**	**.9890**	**.0080**

TABLE VII

ABLATION STUDY ON THE ORDERING OF GC-VCM AND LC-VCM

Configuration	ORSSD Dataset			
	S_α ↑	F_β^{max} ↑	E_ξ^{max} ↑	$\mathcal{M}$ ↓
Serial (GC→LC)	.9515	.9307	.9890	.0080
Serial (LC→GC)	.9501	.9303	.9880	.0082
Parallel (GC ‖ LC)	.9497	.9300	.9886	.0082

being 7.22M/5.79G. This performance degradation indicates that GCTU plays a crucial role in improving global feature transfer and overall representation quality. In contrast, removing VCU yields S_α = 0.9491, F_β^{max} = 0.9222, E_ξ^{max} = 0.9845, and $\mathcal{M}$ = 0.0088, with slightly reduced complexity (7.21M/5.78G). Although the performance drop is smaller than that of removing GCTU, the results still demonstrate that VCU consistently contributes to better feature quality and overall performance.

3) Effectiveness of Each Component in LC-VCM: To quantify the contribution of each component in LC-VCM, we conduct ablation studies by removing LCTU or VCU individually. As shown in Table VI, removing LCTU degrades performance on ORSSD to S_α = 0.9481, F_β^{max} = 0.9236, and E_ξ^{max} = 0.9843, with MAE increasing to 0.0095, indicating that LCTU plays an important role in local feature transfer and fine-detail modeling. Removing VCU also causes performance drops (S_α = 0.9486, F_β^{max} = 0.9244, E_ξ^{max} = 0.9840, and MAE = 0.0092), showing that VCU provides additional gains for local feature quality.

We further study the group number used in feature group fusion. Table VI shows that *Groups = 4* achieves the best overall results (S_α = 0.9515, F_β^{max} = 0.9307, E_ξ^{max} = 0.9890, and MAE = 0.0080), while using fewer groups leads to consistent degradation (*Groups = 1*:MAE = 0.0084 and *Groups = 2*:MAE = 0.0081). The parameter and FLOPs differences across group settings are marginal, suggesting that the gains mainly come from more effective feature interaction rather than increased complexity.

4) Effectiveness of GC-VCM and LC-VCM Ordering: We compare different configurations of the VCMs, including reversing the serial order (LC→GC) and a parallel dual-branch design. As shown in Table VII, the proposed serial GC→LC achieves the best performance across all metrics, with the highest S_α, F_β^{max}, and E_ξ^{max}, and the lowest MAE. Reversing the order (LC→GC) leads to a consistent performance drop, suggesting that applying local compensation before GC regularization may amplify noisy local responses. The parallel design (GC ‖ LC) does not bring further improvement and shows slightly higher MAE, indicating that increased structural complexity does not necessarily improve saliency detection.

Overall, these results support the proposed global-to-local ordering, where GC regularization facilitates more effective subsequent local refinement.

5) Visual Analysis: To gain deeper insight into intercomponent interactions, we conduct a visual comparison under two ablation configurations: removing LC-VCM (*w/o LC-VCM*) and removing GC-VCM (*w/o GC-VCM*), as shown in Fig. 8. For each case, the leftmost column presents the input ORSI, the GT saliency map, and the final saliency map (Result). To the right of the dashed line, the three rows show encoder features from VMamba, output features after removing the respective module, and decoder features, revealing how representations evolve through global and local compensation. Across variants, the VMamba encoder emphasizes global dependencies and cross-scale semantics via state-space modeling, yielding coherent but relatively smooth feature maps at high-frequency boundaries.

In the *w/o LC-VCM* case, the remaining GC-VCM enhances cross-scale alignment and suppresses homogeneous background correlations, enabling reliable localization of major objects; however, the final maps show boundary erosion and missed detections of small targets in clutter, suggesting that GC alignment alone is insufficient to preserve fine, low-contrast boundaries without explicit local boundary/detail cues. This observation implies that the global pathway mainly stabilizes scale-consistent semantics, while recovering thin structures still benefits from explicit local compensation.

Conversely, in the *w/o GC-VCM* case, the retained LC-VCM sharpens edges and preserves fine structures through localized compensation, producing clearer object contours. Yet the lack of GC-VCM weakens global semantic consistency, making predictions appear more susceptible to textured background interference and scale inconsistency: large objects become fragmented, and small targets are spuriously activated or suppressed due to insufficient global context. This indicates that local refinement improves boundary fidelity, but still relies on global cross-scale regularization to maintain scale-consistent objectness under complex backgrounds.

Fig. 9 provides a module-level visualization, where (a) instantiates the global pathway with GC-VCM-3 and (b) instantiates the local pathway with LC-VCM-1. These two modules are selected according to their roles in the overall pipeline shown in Fig. 2. GC-VCM-3 is the first global interaction stage that injects the deepest semantic representation into a higher-resolution feature level, and thus best reflects the initiation of cross-scale information transfer; in contrast, LC-VCM-1 is the closest local compensation module to the final prediction branch (Dec-1 and S_1), operating on the highest-resolution features and therefore most directly affecting boundary/detail reconstruction. As illustrated, each module follows a unified pipeline: the cross-scale interaction unit (GCTU/LCTU) first exchanges information between adjacent-scale features to capture complementary cross-scale cues, and

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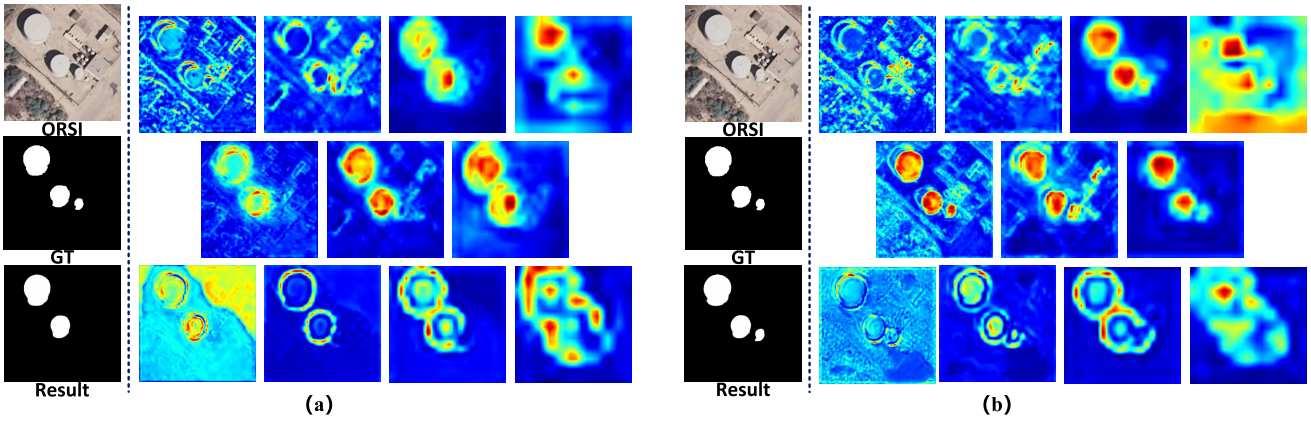


Fig. 8. Visual comparison of two ablation configurations (a) without LC-VCM and (b) without GC-VCM. For each example, the leftmost column shows the input ORSI, the GT saliency map, and the model's saliency map (result). The first row to the right of the dashed line shows the encoder features, the second row shows the output features after removing the corresponding module, and the third row shows the decoder features.

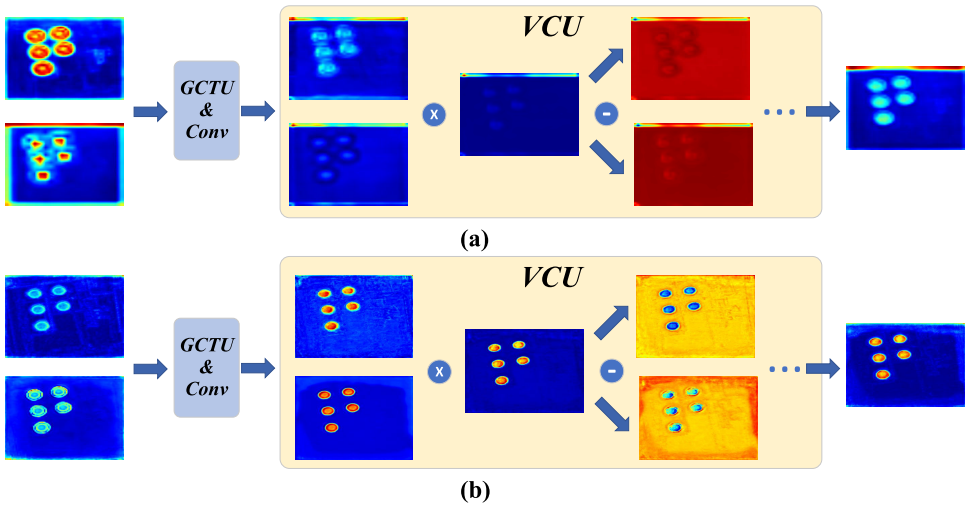


Fig. 9. Visualization of the proposed modules. (a) GC-VCM visualization (GC-VCM-3). (b) LC-VCM visualization (LC-VCM-1).

the subsequent VCU performs variance-aware compensation to refine the fused representations. The elementwise multiplication (⊗) and subtraction (⊖) are used as lightweight interaction operators within VCU to modulate cross-scale responses, which tend to emphasize informative structures while suppressing redundant correlations.

More specifically, in Fig. 9(a), the responses after GC-VCM-3 become more spatially coherent across scales, with reduced responses on repetitive background textures, indicating that the global pathway primarily strengthens scale-consistent semantic aggregation and suppresses redundant correlations during early cross-scale fusion. In Fig. 9(b), the activations after LC-VCM-1 are more concentrated around boundaries and thin structures while keeping interior responses compact, which is consistent with its role in refining high-resolution details immediately before the final decoder prediction. Overall, the visualization highlights the different emphases of the two pathways: GC-VCM-3 focuses on globally consistent cross-scale aggregation and background suppression, whereas LC-VCM-1 concentrates on localized refinement for enhancing boundary-sensitive details, and their complementarity is consistent with the distinct failure modes observed in Fig. 8.

These observations reveal a clear division of labor and complementarity. VMamba and GC-VCM form a robust global pathway: VMamba provides a strong multiscale semantic prior, while GC-VCM enforces cross-scale consistency and suppresses background noise, yielding a clean semantic scaffold. LC-VCM then injects variance-aware local detail and boundary cues, enabling the cascaded convolutional decoder to reconstruct sharp contours without sacrificing spatial fidelity. The qualitative evidence in Figs. 8 and 9 substantiates the synergy among GC-VCM, LC-VCM, the VMamba encoder, and the decoder, providing a mechanistic explanation for the performance gains observed in our quantitative ablation studies.

6) Number of Channels: We conducted experiments to verify why the normalized channel number was set to 32. As shown in Table VIII, when the number of channels increases from 32 to 64 and 128, the performance on some metrics (e.g., S_α and E_ξ^{max}) shows slight improvements. However, these gains are very limited compared to the substantial growth

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TABLE VIII

ABLATION EXPERIMENT ON THE NORMALIZED CHANNEL NUMBER OF FEATURES IN MAMBACVC. THE BEST ONE IN EACH COLUMN IS IN BOLD

No.	Number of channels	#Param $(M)\downarrow$	FLOPs $(G)\downarrow$	ORSSD Dataset			
				$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$
1	**32**	**7.26**	**6.25**	.9515	.9307	.9890	**.0080**
2	64	8.79	13.99	.9516	.9314	.9867	.0085
3	128	13.76	43.00	**.9529**	**.9318**	**.9893**	.0083

TABLE IX

ABLATION EXPERIMENT OF MAMBACVC'S INPUT SIZE. THE BEST RESULT IN EACH COLUMN IS IN BOLD

No.	Input Size	#Param $(M)\downarrow$	FLOPs $(G)\downarrow$	ORSSD Dataset			
				$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$
1	$\mathbf{224\times224}$	**7.26**	**6.25**	.9515	**.9307**	**.9890**	**.0080**
2	256×256	7.26	8.16	.9517	.9291	.9863	.0090
3	288×288	7.26	10.33	**.9522**	.9278	.9862	.0088

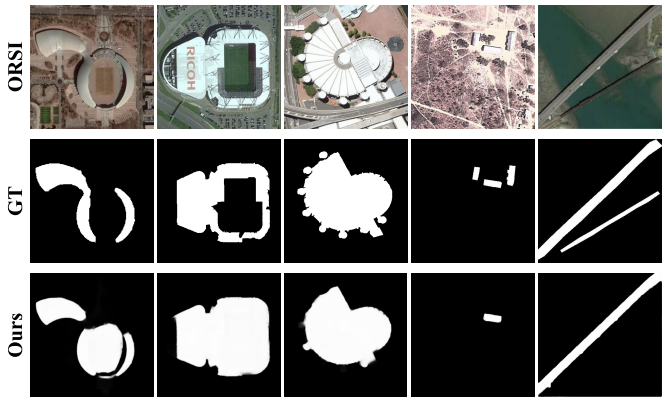


Fig. 10. Several representative failure cases of our model.

in model complexity, where the FLOPs increase from 6.25G to 43.00G. In contrast, the configuration with 32 channels achieves comparable or even superior results, especially on the $\mathcal{M}$ metric. This indicates that setting the normalized channel number to 32 not only minimizes the parameter size (7.26M) and computational cost but also ensures competitive accuracy.

7) Input Size: We conducted experiments to evaluate the impact of input image size on model performance. As shown in Table IX, we tested three cases: 224 × 224 (No. 1), 256 × 256 (No. 2), and 288 × 288 (No. 3). The results in Table IX show that the model's performance in the other two cases slightly degrades compared to the baseline (224×224). This indicates that image input size has little impact on MambaCVC's performance, demonstrating the robustness of the proposed method.

D. Discussion

As shown in Fig. 10, several representative failure cases reveal the current limitations of our model. The first two examples contain hollow salient objects, where the inner empty regions are incorrectly filled. This occurs because the current variance compensation mechanism emphasizes GC consistency while neglecting subtle intraobject variance cues. The third example involves a large object with fine and irregular boundary structures, where the predicted mask fails to preserve detailed contours, resulting in slightly blurred edges. The fourth case shows a small-scale target, which is partially missed due to weak feature responses during multiscale fusion. In the fifth case, which includes a narrow and elongated object, our model exhibits mild oversmoothing along the object boundary, reducing edge precision.

From a system-level perspective, MambaCVC still inherits certain architectural constraints from the Mamba backbone. Although Mamba achieves efficient sequence modeling through selective state updates, its sequential dependency along the sequence dimension restricts full parallelization. In addition, our dual VCMs are designed in a serial manner, leading to strong dependence on preceding outputs and slower inference speed.

In future work, we plan to optimize the Mamba backbone for improved parallel efficiency and redesign the compensation modules into a partially parallel structure to reduce serial latency. These enhancements are expected to further improve real-time performance and enhance the model's ability to handle structurally complex or fine-grained targets.

V. CONCLUSION

This article introduces MambaCVC, a lightweight network that effectively addresses the challenges of ORSI-SOD through a principled dual-correlation variance compensation strategy. By leveraging the linear complexity advantages of the VMamba and introducing theoretically grounded GC-VCM and LC-VCM, MambaCVC successfully bridges the gap between computational efficiency and detection accuracy in complex remote sensing scenarios. The correlation variance compensation explicitly models interlayer feature statistics to enhance discriminative learning, with GC-VCM enabling effective global context modeling across scales and LC-VCM preserving critical fine-grained details essential for accurate boundary delineation. This dual-compensation approach proves particularly effective for handling the extreme scale variations and cluttered backgrounds characteristic of remote sensing imagery. Experimental results on three public ORSI-SOD datasets demonstrate that MambaCVC achieves state-of-the-art performance with only 6.25G FLOPs and 7.26M parameters, establishing superior efficiency–accuracy tradeoffs for resource-constrained remote sensing applications.

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